HUMAN ANATOMY & PHYSIOLOGY-II

UNIT 2 NOTES

DIGESTIVE SYSTEM

- ANATOMY OF GI TRACT
- REGULATION OF ACID PRODUCTION
- SALIVARY GLANDS
- LIVER & PANCREAS

ENERGETICS

- FORMATION & ROLE OF ATP CREATININE PHOSPHATE & BMR

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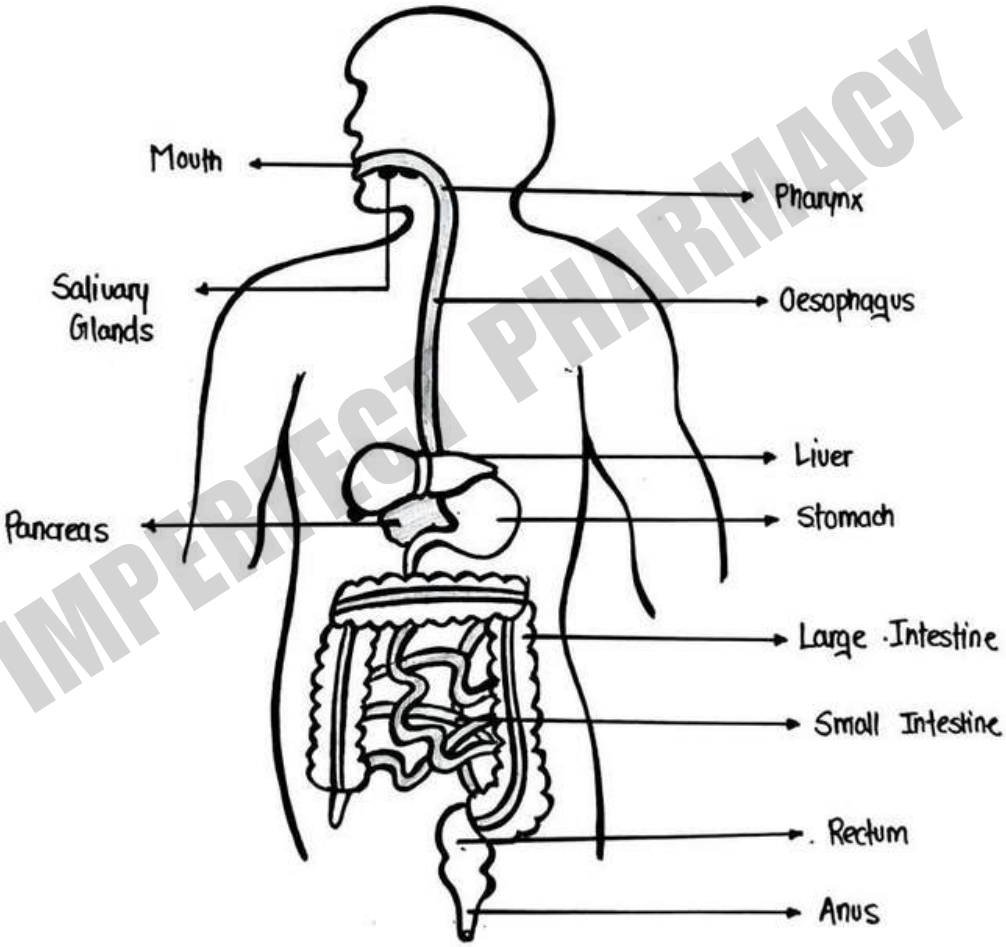
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DIGESTIVE SYSTEM

- The major objective of eating food is to get nutrients.
- The conversion of complex food into simple & absorable form is known as digestion & the system that performs the process of digestion is known as digestive system.
- Digestive system simply breakdown the food into nutrients.



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Stages Of Digestion

- Ingestion : Receiving of food
- Propulsion : Movement of food
- Digestion : Mechanical & chemical breakdown
- Absorbtion : Absorbtion of nutrients
- Excretion : Removal / Excretion of food

Parts of Digestive System

- The digestive is mainly consist of Gastrointestinal Tract & Digestive Glands also known as Accessory Glands.
- The major organs of GIT includes :

(1) Mouth
(2) Pharynx
(3) Oesophagus
(4) Stomach
(5) Small Intestine
(6) Large Intestine
(7) Rectum
(8) Anus

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① MOUTH

- It is also known as Oral cavity or buccal cavity.
- It is the first part of gastrointestinal tract.
- It is surrounded by upper & lower lips.
- The mouth mainly consist of lips, cheeks, gum, teeth, salivary glands & Tongue.
- The space b/w gums & cheeks is known as Vestibule
- The roof boccal cavity is known as Palate.
- Palate are of two types : Soft Palate & Hard Palate.

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Teeth :
- Teeth mainly perform mechanical digestion.
- They breakdown the food into smaller pieces
- There are total 32 teeth in mouth

① Incisors : 8
② Canines : 4
③ Premolars : 8
④ Molars : 12

Tongue :
- The tongue lies on floor of mouth
- It is attached with Hyoid bone
- It mixes saliva into food & convert into bolus
- It contains taste receptors.

① Sweet
② Salty
③ To Sour
④ Bitter

② PHARYNX

- It is a muscular tube like structure
- It lies blw mouth & oesophagus.
- It is the common pathway for digestive & respiratory tract.
- It is also known as Throat.
- It passes the bolus into oesophagus
- Epiglottis make sure the pathway of food (bolus) from pharynx to oesophagus.
- It is further divided into three parts.

(1) Nasopharynx
(2) Oropharynx
(3) Laryngopharynx

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③ OESOPHAGUS

- It is also known as food pipe.
- It is a hollow tube like structure that carries food (bolus) from pharynx to stomach.
- It lies b/w trachea in front & vertebral column at back.
- No digestion takes place in oesophagus as digestive enzymes are absent.
- Oesophagus opens into stomach

④ STOMACH

- It is a J shaped pouched like structure.
- It lies in the left side of abdominal cavity.
- Capacity : 30 ml - 1.5 Litre.
- It secretes digestive fluids called Gastric Juice.
- It receives food from oesophagus.
- It secretes gastric juice into bolus & convert into Chyme.

Wall of Stomach

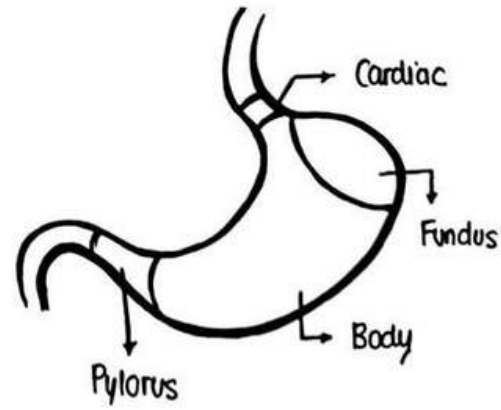
The wall of stomach is made of 4 layers :

① Serosa
② Masculanis
③ Submucosa
④ Mucosa

Parts of Stomach

It can be divided into 4 parts

① Cardiac
② Fundus
③ Body
④ Pylorus



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Cardiac :
- It is the uppermost part of stomach.
- It is attached with lower end of oesophagus.
- It contains cardiac spinchter.
- It prevents backflow of food from stomach to oesophagus otherwise it can cause heartburn

Fundus :
- It is also the upper part of stomach.
- Fundus contains special types of cells known as Chief cells & Parietal Cells.
- CHIEF CELLS : It produces various enzymes includes-
  1. Pepsin
  2. Renin
  3. Lipase
- PARIETAL CELLS : It mainly produces HCl

Body :
- It lies in the central region of stomach.
- It is the largest part of stomach
- It digest protein & stores food until it reaches into small intestine

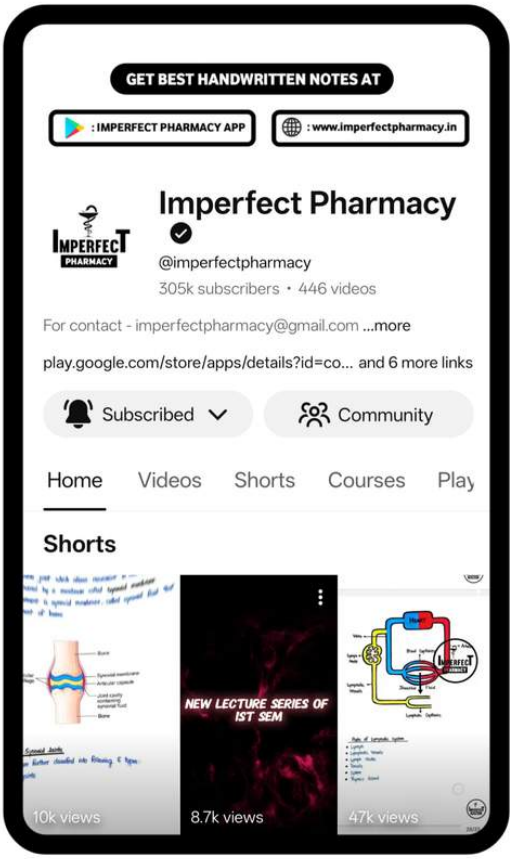
Pylorus :
- It is the lower part of stomach.
- It is connected with duodenum of small intestine.
- It transfers the food from stomach to small intestine.
- It contains special types of cells known as G-CELLS.
- G cells Secretes Gastrin Harmone.

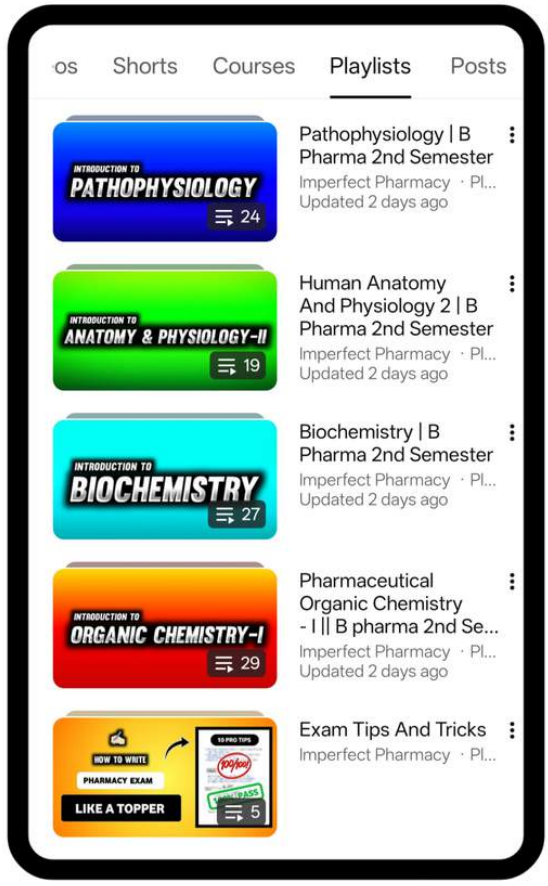
NOTE : In the complete epithelium of stomach there are so many goblet cells are present that secretes mucous & it protects stomach from HCl.

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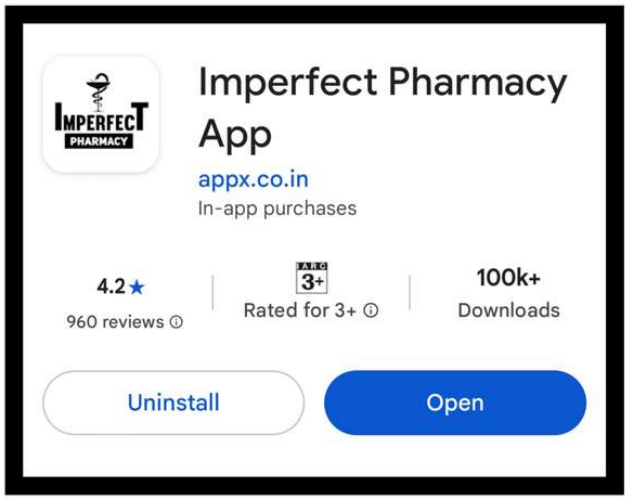
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Role of Gastric HCl

- The HCl (Hydrochloric acid) plays major role in digestion of food in the stomach.
- It is secreted by parietal cells of stomach.
- It helps to convert bolus into chyme.
- It helps in the breakdown of complex food.
- It converts pepsinogen into pepsin that helps in protein digestion.
- It also shows antiseptic action by killing bacteria.

Functions of Stomach

- It stores the food for a long time & after that transferred into small intestine slowly. and it provides enough time for digestion & absorbtion of food in small intestine.
- Storage Time :

(i) Fat : 6-8 hours
(ii) Proteins : 4-5 hours
(iii) Carbohydrates : 1-2 hours

- It helps in the chemical breakdown of food.
- Gastric juice mainly helps in the digestion of protein.
- It secretes mucous that protects the stomach from HCl.
- It kills bacteria & other dust particles from food.

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⑤ SMALL INTESTINE

- Small intestine is the site of final digestion of food.
- It is a long & narrow tube whose upper part is attached to
- stomach while lower part is attached with large intestine.
- It is about 6-7 meter long.
- It is the site of major absorbtion & digestion
- It is further divided into three parts

① Duodenum
② Jejunum
③ Ileum

Duodenum :

- Duodenum accepts only Alkaline food.
- It is shortest & widest part of small intestine.
- It is approx 10-11 inches long.
- It receives bile juice, pancreatic juice & food from liver, pancreas & stomach respectively.
- Secretions of Duodenum :

(i) Enterogastrone
(ii) Secretin
(iii) Pancreoenzyme
(iv) Cholecystokinin

Jejunum

- It is about 4-8 feet long.
- It is the major site for absorbtion of sugars, amino acids & fatty acids.
- Final breakdown of food completed here.
- It secretes intestinal juice.

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ILEUM

- It is last part of small intestine connected with the cecum (first part of large intestine).
- It is about 10- 11 feet long.
- It is the largest section of small intestine.
- It is the major site of absorbtion of nutrients, bile, vitamins etc.

Enzymes Responsible For Digestion in ~~Protein~~

- Enterokinase : Converts trypsinogen into trypsin
- Erepsin : Converts polypeptides into amino acids.
- Sucrase, Maltase & Lactase : Converts disaccharides into monosaccharids.

Absorbtion in Small Intestine

- The absorbtion in small intestine mainly occurs through Villi & Microvilli
- Villi : Finger like projections
- Microvilli : Tiny hair like structures on Villi

⑥ LARGE INTESTINE

- It starts from ileum & ends on rectum.
- It is approx 1.5 m long.
- It mainly absorbs water from digested food.
- It is further divided into 5 parts.

① Cecum
② Ascending colon
③ Descending Colon
④ Transverse Colon
⑤ Sigmoid Colon

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⑦ RECTUM

- It stores the remaining waste of Food.
- It opens outside through anus.
- The undigested or unabsorbed substances that is stored in the anus are called Faeces.

⑧ ANUS

- It is the final portion of the Gastrointestinal Tract.
- The undigested and unabsorbed food that is simply known as
- Faeces excreted out from body through anus & it is a voluntary process.
- Two types of spinchter muscles found in anus.

① Internal : Involuntary
② External : Voluntary

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DIGESTIVE GLANDS

- Digestive glands are the accessory glands of digestive system that helps in the process of digestion.
- They mainly includes :

① Salivary Glands
② Liver
③ Gall Bladder
④ Pancreas

① SALIVARY GLANDS

- Salivary glands are the exocrine glands present in the mouth that secretes saliva in GIT
- Saliva mixes with food & convert it into Bolus.
- Human salivary glands produces 0.5 L to 1.5 L saliva daily.
- Saliva contains digestive enzymes mainly Salivary Amylase which helps in the carbohydrate digestion.
- It also have antibacterial effect of fighting off bacteria.

Saliva contains 99.5 % water & 0.5% proteins & mineral salts.

Types of salivary glands

There are mainly 3 types of salivary glands present in humans :

① Parotid Glands
② Submandibular Glands
③ Sublingual Glands.

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Parotid Glands

- These are the largest salivary glands.
- They are located below the auditory capsule ear
- Duct : Stenson's Duct.
- It opens in upper jaw.
- It contributes 25% saliva.
- It secretes serous type saliva

Submandibular Glands

- They are second largest salivary glands (smaller than parotid & bigger than sublingual).
- It opens in the buccal cavity.
- Duct : Wharton's Duct.
- It makes 70% of total saliva
- It secretes both serous & mucous saliva

Sublingual Glands

- They are the smallest salivary glands.
- They are located below the tongue.
- It makes 5% of total saliva.
- Duct : Bartholin's Duct.
- They don't contain enzymes.

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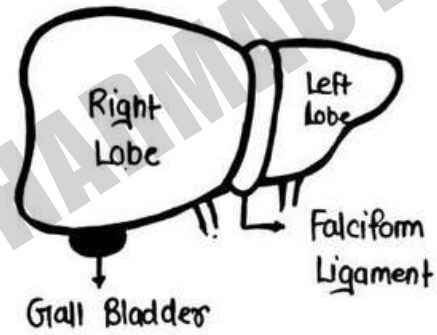
② LIVER

- Liver is the largest gland or largest organ of our body.
- Its weight is about 1.5 kg.
- It has highest regeneration power
- Liver performs maximum number of functions in our body.
- It contains lots of mitochondria.
- It is reddish brown - colour in human

Surfaces of Liver

Liver have four surfaces :

① Superior Surface
② Inferior Surface
③ Anterior Surface
④ Posterior Surface



Lobes of Liver

- Liver contains two lobes

① Right Lobe
② Left Lobe

- Right lobe is much larger than left lobe.
- Right lobe forms 5/6th part while left lobe forms 1/6th part
- Right & Left lobes are separated by Falciform Ligaments.
- Right duct arises from right lobe & Left duct arises from left lobe of liver.
- Both right & left ducts combines & form a common hepatic duct

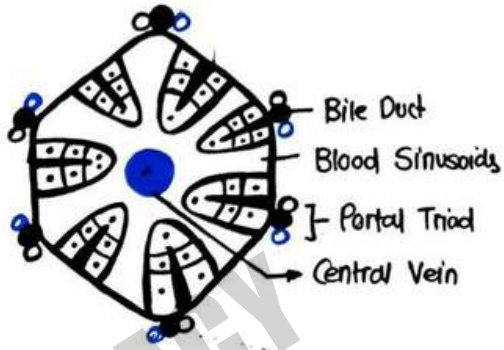
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Internal Structure of Liver

- The liver is made up of small hexagonal units present in large number in liver.
- Each lobule has a central vein or intralobular vein surrounded by several portal Triads
- The portal triad is consist of

① Portal Vein
② Bile duct
③ Hepatic Artery



Functions of Liver

- Secretion of bile juice
- Synthesis & storage of Glycogen
- Synthesis of Urea
- Metabolism of carbohydrates, fats & Proteins.
- Vitamins & Minerals Storage.
- Synthesis of Heparin
- Blood-sugar regulation
- Detoxification & Purification of blood.

③ GALL BLADDER

- Gall bladder lies below the right lobe
- It stores & concentrates the bile from liver.
- Gall bladder contains cystic duct
- Microvilli on gall bladder absorbs moisture from bile & makes it concentrated

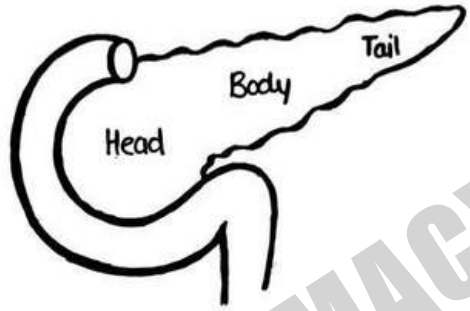
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④ PANCREAS

- It is a mixed gland
- It is located behind the stomach in upper left abdomen.
- It secretes both enzymes & harmones
- It can be divided into three parts :

① ~~Lipase~~ Head
② ~~Amylase~~ Body
③ ~~Trypsin~~ Tail.



Enzymes Secretion by Pancreas

Pancreatic Juice mainly contains :

- Lipase : Converts Fatty ~~acid~~ into Fatty acids & glycerols.
- Amylase : Converts Starch into Maltase
- Trypsin : Converts Peptones into Amino acids.

Harmones Secretion by Pancreas

Islets of Langerhans contains mainly 3 types of cells that secretes following harmones.

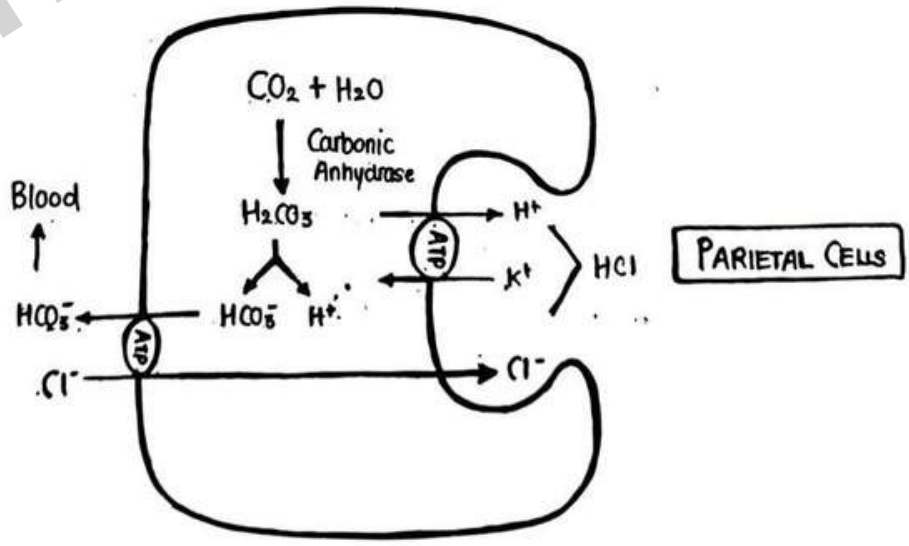
- Alpha Cells : Glucagon.
- Beta cells : Insulin
- Delta Cells : Somatostatin

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Acid Production In Stomach

- HCl is produced by the parietal cells of stomach
- Hydrochloric acid secretion is an active process that takes place in the canaliculi of parietal cells.
- Carbon di oxide is derived from metabolic activities of parietal cells.
- The carbon di oxide is present in parietal cells combines with H_2O & form carbonic acid (H_2CO_3) in the presence of carbonic anhydrase.
- The enzyme carbonic anhydrase present in high concentration in parietal cells.
- Carbonic acid is most unstable compound & hence it immediately splits into hydrogen ion (H^+) & bicarbonate ion (HCO_3^-)
- The hydrogen ion is actively pumped into canaliculus of parietal cells. through K^+/H^+ Atpase pump & the energy for this process is derived from glucose
- Now simultaneously along with this process Cl^- ions also pumped into canaliculus of parietal cells.
- Chloride (Cl^-) is derived from NaCl in blood.
- Now in the final step H^+ combines with Cl^- & form HCl

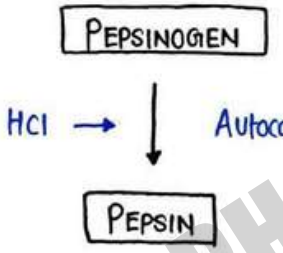


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ROLE OF PEPSIN IN PROTEIN DIGESTION

- Pepsin is the enzyme secreted by chief cells of stomach
- It is the only proteolytic enzyme present in Gastric Juice (proteolytic enzymes are those that helps in breakdown or digestion of protein.
- The chief cells produce pepsin in the inactivated form known as Pepsinogen.
- Now this pepsinogen activates in the form of Pepsin with the help of HCl secreted by parietal cells.



- After activating in the form of pepsin it breakdowns the proteins into smaller Peptides & Amino acids that can be easily absorbed in small intestine.
- Although pepsin plays major role in protein digestion but Most of the protein are digested ~~by~~ in the duodenum & jejunum of small intestine by pancreatic juice

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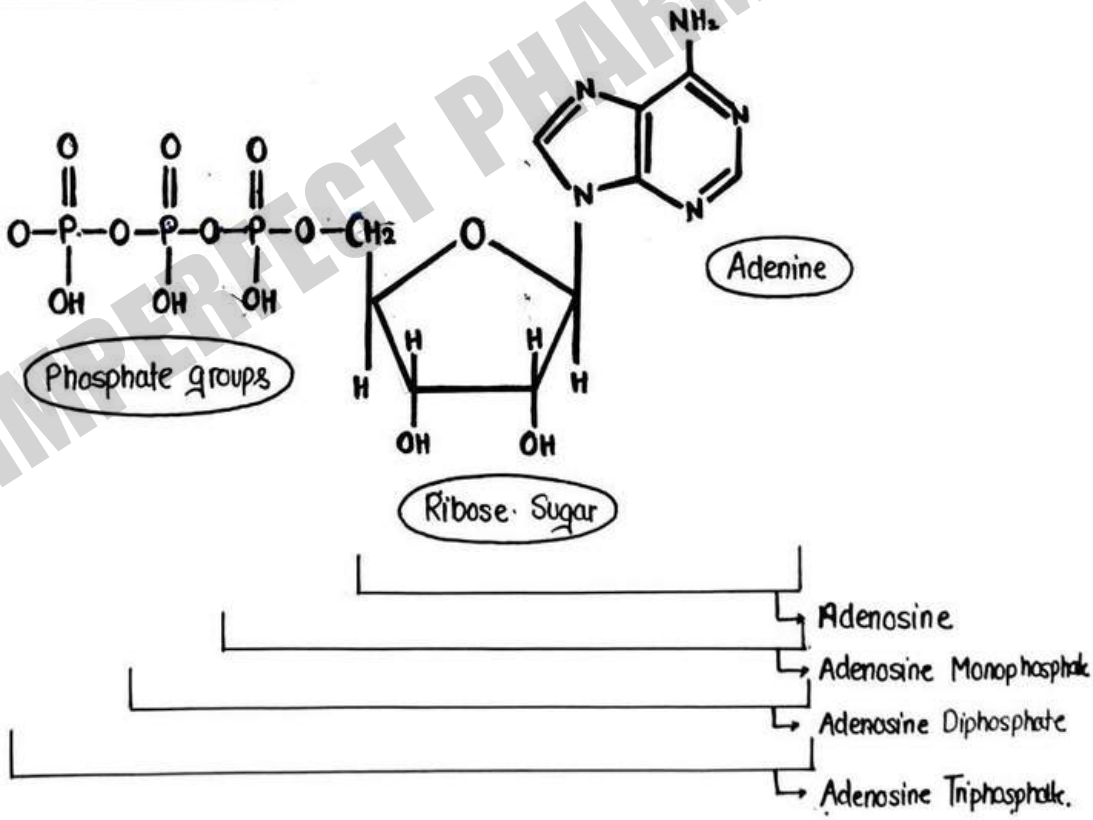
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ATP

- The full form of ATP is Adenosine Tri Phosphate.
- It is a multifunctional nucleotide coenzyme.
- It is the most important form of chemical energy in all cells.
- It is also known as Energy currency of cell.
- It is produced by phosphorylation and cellular respiration.
- It is used by enzymes and structural proteins in many cellular process.
- One molecule of ATP contains 3 phosphate groups.

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Structure of ATP

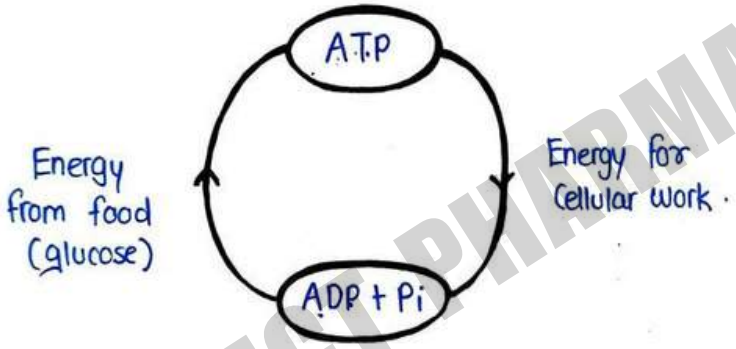


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- The energy from ATP is released by breaking of high energy phosphoanhydride bonds via hydrolysis.

- $ATP + H_2O \rightarrow ADP + Pi$
- $ADP + H_2O \rightarrow AMP + Pi$
- $ATP + H_2O \rightarrow AMP + PPi$

ATP- ADP Cycle



Biological Significance / Functions of ATP

- ATP is the biggest source of energy within all living organism.
- It is used in the muscle contraction.
- It is used in active transport across cell membrane.
- It is used in impulse formation.
- It is used in the metabollic processess.
- It is the precursor of C- AMP.

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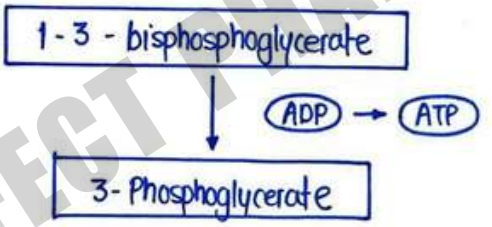
Mechanism Of ATP formation

There are mainly 3 types of mechanism, through which organism form ATP.

① Substrate - Level Phosphorylation
② Oxidative Phosphorylation
③ Photophosphorylation

• Substrate Level Phosphorylation

* This mechanism involved direct transfer of phosphate group from substrate to ADP (adenosine diphosphate), forming ATP.
* This process mainly occurs during Glycolysis & Citric Acid Cycle (Kreb's Cycle) in cellular respiration.
* Example :



• Oxidative Phosphorylation

* This mechanism occurs in the mitochondria & involves the use of energy derived from electron transport chain to phosphorylate ADP to ATP.
* It is the major mechanism for ATP synthesis in Aerobic Respiration.

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- **Photophosphorylation**

  It occurs only in chlorophyll containing plant cells or in certain bacteria that contain other light absorbing pigments.

Role Of ATP

- ATP plays a crucial role in the cell as primary energy carrier and universal energy currency. its roles can be summarized as follows :
- ATP stores & releases energy in its phosphate bonds, when ATP booken down into ADP & Pi, energy is released that powers various, cellular process.
- ATP provides necessary energy to synthesize macromolecules such as Proteins, Lipids, Nucleic Acids etc.
- ATP is involved in Active Transport Processes through cell membrane.
- It powers mechanical works in cells, including muscle contraction, movement of organelles within cells & cellular motility.
- ATP regulates & controls various metabolic pathways.

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METABOLISM

- Metabolism refers to complex set of biochemical reactions that occur within living organism to sustain life.
- These reactions collectively enable the organism to acquire, transform, store & utilize energy & nutrients essential for growth, reproduction, maintainence & response to the environment.
- It can be further divided into 2 types :

① Catabolism
② Anabolism

Catabolism

Catabolism is simply defined as breakdown of complex molecules into simpler molecules.

Anabolism

Anabolism is simply defined as synthesis of complex molecules from simpler ones.

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BMR

- BMR stands for Basal Metabolic Rate.
- It refers to the amount of energy needed by an individual in awake state at rest to maintain vital functioning of body i.e., respiration, circulation, Nerve functions, kidney functions, ion transport across membrane & other cellular activities in post absorption phase.
- Normal Value : In Males : 35-38 Cal / sq-m / hr
  In Females : 32-35 Cal / sq-m / hr

Determination of BMR

While determining BMR, we should take care that person must :

- Awake
- Rest
- Post - Absorption Phase
- Neutral Temperature
- Mental Rest

There are two methods for determination of BMR :

① Direct Method
② Indirect Method

Factors Affecting BMR

- Grender
- Age
- Physical Activity
- Hormones
- Stress

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